

Input Ontology (IO)

Score: 3/5 — Moderate gaps | Confidence: high

Benchmark: WXIMPACTBENCH | Context: Argentine Government Disaster Reporting Reliability — Policy Staff

Input Ontology definition: Whether the benchmark’s test case categories cover the query types expected in deployment.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The six top-level impact categories (Infrastructural, Political, Financial, Ecological, Agricultural, Human Health) are broadly applicable to Argentine disaster types and the elicitation confirms they 'could accommodate the named disaster types' [A1]. Empirical inspection confirms each category appears in the data with positive examples [DATASET-D14, D16, D18]. However, the ontology is constructed entirely around formal newspaper articles [Q18, Q115], with no task type targeting social media, press releases, or short-form informal content — the deployment's highest-priority input channels. No sub-labels exist within the six categories, a recognized gap for Argentine policy granularity (e.g., villa miseria vs. formal infrastructure). The chunked dataset also includes a high proportion (~65%) of off-topic zero-label fragments [DATASET-D1, D2, D7, D23] that may dilute the discriminative validity of the ontology for genuine disaster classification. The QA task is closed-retrieval only [Q89], precluding the open-retrieval setting most relevant to a deployment scanning real source streams.

ID	Evidence
Q18	"Climate impact analysis (Thulke et al., 2024) aims to help society make correct decisions about climate-related challenges affecting communities, e.g., understanding the weather impacts on society." (p.2)
Q89	"The practical open-retrieval setting, i.e., identifying the relevant articles from a huge database, is left for future studies, which could further facilitate knowledge enhancement in understanding disruptive weather impacts." (p.9)
Q115	"Table 16 presents the prompt designed to analyze historical newspaper texts and classify them into six distinct impact categories based on explicit mentions of weather-related events." (p.16)
D1	John's Cloudy -4 -4 United States Max Min Atlanta Cloudy 17 4 Boston Cloudy 3 -6 Chicago Cloudy 2 2 Dallas Thunderstorms 11 2... — Weather forecast table with temperature/condition data; no impact content
D2	But the real beneficiaries of the work-at-home trend are the employees themselves... She takes a turn at the table in a lay way (5) 29, A good one is of assistance (9) Down 1, Full of ill-will and disease (9)... — Non-weather content: work-from-home editorial + crossword puzzle clues
D7	National Lampoon's Van Wilder 2: The Rise of Taj Starring: Kal Penn, Lauren Cohan and Daniel Percival Playing at: AMC, Brossard, Cavendish, Colossus, LaSalle cinemas. — Film review; no weather content at all
D14	by six o'clock the electric car services in all the important points west of Toronto had been completely paralyzed... an unprecedented snowfalls and heavy winds. — 1894 blizzard; infrastructure (transit) disruption labeled; North American geography
D16	Aid Laurent said that there were mountains of snow all over the city. Already \$30,000 has been expended in removing it... Seven hundred and fifty men and 450 carters were employed... — 1887 Montreal snow removal debate in city council; all three labeled labels are plausible
D18	more than 100,000 homes damaged and crops on about 175,000 hectares of farmland destroyed... death toll in Chongqing in China's southwest rose to 42 people and 12 missing from torrential downpours — 2007 global flood roundup (England, China, Pakistan) — non-Canadian events used as positive examples
D23	-2 City res y 5000 5 5 5 Aiexavn 8000 50 50 50 Ckearcdnf 835 825 25..." / "Escenvt 115780 298 J80 289 19 Caiunoil 8000 J9 J9 29... — OCR-corrupted stock market tables; repeated for same article id across multiple rows

Strengths

- Six top-level categories map cleanly onto Argentine disaster impact types per elicitation [A1, Q11]
- All six labels are empirically present and distinguishable in the data, with both single-label and multi-label positive cases [DATASET-D14, D16, D18]

- Multi-label simultaneous classification design [Q57] matches realistic disaster reports that span multiple impact dimensions

ID	Evidence
Q11	"The articles are selected by topic modeling, including six impact categories (infrastructural, political, financial, ecological, agricultural, and human health), which are informed by previous studies (Imran et al., 2016a) and align with modern disaster impact assessment frameworks (Silva et al., 2022)." (p.2)
Q57	"Different from traditional methods that decompose multi-label text classification into multiple binary classification tasks (Boutell et al., 2004; Liu et al., 2017), we simultaneously identify all relevant disruptive weather impacts for each input by calling the LLM once." (p.6)
D14	by six o'clock the electric car services in all the important points west of Toronto had been completely paralyzed... an unprecedented snowfalls and heavy winds. — 1894 blizzard; infrastructure (transit) disruption labeled; North American geography
D16	Aid Laurent said that there were mountains of snow all over the city. Already \$30,000 has been expended in removing it... Seven hundred and fifty men and 450 carters were employed... — 1887 Montreal snow removal debate in city council; all three labeled labels are plausible
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Checklist

Checklist item definitions:

ID	Assessment Question
IO-1	Identify test case categories required for regional deployment
IO-2	Check if taxonomy omits regionally relevant categories
IO-3	Check if taxonomy includes irrelevant categories
IO-4	Document category gaps harming content validity

Checklist responses:

IO-1: Argentine deployment requires categories covering: urban flash floods (AMBA), sudestada coastal flooding, pampero windstorms, ENSO-driven Pampa drought, Patagonian wildfire, NOA flood/drought, plus credibility/reliability scoring across formal press, social media, and informal community channels [WEB-1, WEB-5, WEB-7].

IO-2: Benchmark omits: (a) any task targeting social media or short-form informal content [Q18, Q115], (b) credibility/misinformation/source-reliability scoring (the deployment's core task per A3), (c) sub-labels within the six categories needed for Argentine policy granularity [A1], (d) open-retrieval QA setting [Q89], and (e) Spanish-language task framing.

IO-3: No categories appear actively irrelevant to the Argentine context — the six top-level categories are broadly applicable per A1. However, the LDA-derived 15 weather event types [Q101] are not surfaced as evaluation categories, and the chunked dataset contains a high proportion of off-topic content (crosswords, film reviews, stock tables) labeled all-zero [DATASET-D2, D7, D23], which dilutes the ontology's discriminative signal.

IO-4: Major gaps: (1) no social media / short-form input task type despite this being a HIGH-priority deployment channel; (2) no credibility scoring task; (3) no sub-labels for Argentine-specific granularity (informal settlements, agricultural commodity-specific loss); (4) closed-retrieval QA only.

ID	Evidence
Q18	"Climate impact analysis (Thulke et al., 2024) aims to help society make correct decisions about climate-related challenges affecting communities, e.g., understanding the weather impacts on society." (p.2)
Q89	"The practical open-retrieval setting, i.e., identifying the relevant articles from a huge database, is left for future studies, which could further facilitate knowledge enhancement in understanding disruptive weather impacts." (p.9)
Q101	"Using Latent Dirichlet Allocation, the dataset was categorized into 15 primary weather event types." (p.14)
Q115	"Table 16 presents the prompt designed to analyze historical newspaper texts and classify them into six distinct impact categories based on explicit mentions of weather-related events." (p.16)

D2	But the real beneficiaries of the work-at-home trend are the employees themselves... She takes a turn at the table in a lay way (5) 29, A good one is of assistance (9) Down 1, Full of ill-will and disease (9)... — Non-weather content: work-from-home editorial + crossword puzzle clues
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WEB-1	SINAGIR/AFE Argentine institutional framework operates with different political accountability structure than Canadian framework (https://www.argentina.gob.ar/sinagir/institucional)
WEB-5	98.4% Argentine mobile broadband indicates short-form mobile content dominant in deployment (https://datareportal.com/reports/digital-2026-argentina)
WEB-7	Rural-urban digital divide means disaster-zone content carries different noise characteristics than urban Buenos Aires (https://freedomhouse.org/country/argentina/freedom-net/2024)

Information Gaps

- Whether sub-label schema for villa miseria infrastructure damage could be added without re-annotating the corpus
- Whether credibility scoring could be derived from existing label outputs without dedicated task design

Requires Expert Verification

- Which Argentine sub-categories within each top-level label matter most for SINAGIR/AFE policy decision-making

Remediation

Gap 1: No task type for social media or short-form informal content; closed-retrieval QA only [Q18, Q89]

Recommendation: Augment evaluation with an open-retrieval QA task and a short-form social-media classification task using Argentine Twitter/X disaster content. Treat WXImpactBench's closed-retrieval scores as an upper-bound proxy and supplement with open-retrieval evaluation against an Argentine source corpus.

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Q89	"The practical open-retrieval setting, i.e., identifying the relevant articles from a huge database, is left for future studies, which could further facilitate knowledge enhancement in understanding disruptive weather impacts." (p.9)